

BS-CH101 / BS-CH201 — Chemistry-I

Semester I/II • 3L:1T:0P • 4 Credits

Module 2 Exam Notes: Spectroscopic Techniques and Applications

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

Contents

1	Principles of Spectroscopy and Selection Rules	4
1.1	Electromagnetic Radiation and Its Quantum Nature	4
1.2	The Electromagnetic Spectrum and Spectroscopic Regions	4
1.3	Absorption, Emission and Scattering	5
1.4	Resonance Condition and the Bohr Frequency Rule	5
1.5	Selection Rules: Allowed and Forbidden Transitions	5
1.6	Population of Energy Levels: Boltzmann Distribution	6
1.7	Resolution and Linewidth: The Heisenberg Limit	7
1.8	Transmittance vs Absorbance	7
1.9	Einstein A and B Coefficients	7
1.10	Franck–Condon Principle	8
2	Electronic (UV-Visible) Spectroscopy	9
2.1	Types of Electronic Transitions	9
2.2	Beer–Lambert Law: The Quantitative Backbone	10
2.3	Chromophores and Auxochromes	10
2.4	Effect of Conjugation	11
2.5	Solvent Effects on Electronic Spectra	11
2.6	Block Diagram of a UV-Visible Spectrophotometer	11
2.7	Applications of UV-Visible Spectroscopy	12
2.8	Woodward–Fieser Rules	12
2.9	Isosbestic Point	13
2.10	Single-Beam vs Double-Beam Instruments	13
2.11	Sample Cell (Cuvette) Materials	13
3	Fluorescence and Its Applications in Medicine	14
3.1	Singlet, Triplet and the Jablonski Diagram	14
3.2	Fluorescence vs Phosphorescence	15
3.3	Stokes Shift and Mirror-Image Rule	15
3.4	Quantum Yield and Quenching	16
3.5	Factors Influencing Fluorescence Intensity	16
3.6	Spectrofluorimeter (Block Diagram)	16
3.7	Applications in Medicine and Biology	17

3.8	Kasha's Rule	18
3.9	Lifetime, Quantum Yield, and Radiative Rate	18
3.10	Förster Resonance Energy Transfer (FRET)	19
4	Vibrational and Rotational Spectroscopy of Diatomic Molecules	20
4.1	Rotational Spectroscopy: The Rigid-Rotor Model	20
4.2	Vibrational Spectroscopy: Simple Harmonic Oscillator Model	21
4.3	Vibration–Rotation (Rovibrational) Spectra of Diatomics	22
4.4	Modes of Vibration in Polyatomic Molecules	22
4.5	Characteristic IR Group Frequencies	23
4.6	Block Diagram of an FT-IR Spectrometer	23
4.7	Applications of IR and Microwave Spectroscopy	23
4.8	Centrifugal Distortion of the Rigid Rotor	23
4.9	Isotope Effect on Vibrational and Rotational Spectra	24
4.10	Fingerprint Region vs Functional-Group Region	24
4.11	Sample Preparation for IR Spectroscopy	24
4.12	IR vs Raman: Side-by-Side Comparison	25
5	Nuclear Magnetic Resonance (NMR) and Magnetic Resonance Imaging (MRI)	26
5.1	Nuclear Spin and Magnetic Moment	26
5.2	Chemical Shift	26
5.3	Spin–Spin Coupling and Multiplicity	27
5.4	Block Diagram of an NMR Spectrometer	28
5.5	Applications of NMR	28
5.6	Magnetic Resonance Imaging (MRI)	28
5.7	T_1 and T_2 Contrast	29
5.8	Advantages and Limitations of MRI	29
5.9	Medical Applications of MRI	29
5.10	Origin of Chemical Shift: Shielding Mechanisms	30
5.11	Coupling Constant J and Integration	30
5.12	MRI Contrast Agents	31
6	Surface Characterisation Techniques	32
6.1	Why a Special Technique for Surfaces?	32
6.2	Scanning Electron Microscopy (SEM)	33
6.3	X-ray Photoelectron Spectroscopy (XPS / ESCA)	33
6.4	Atomic Force Microscopy (AFM)	33
6.5	Comparison of Major Surface Techniques	34
6.6	TEM vs SEM	34
6.7	Scanning Tunnelling Microscopy (STM)	35
6.8	BET Surface Area Analysis	35
6.9	Contact-Angle Measurement: Surface Wettability	36

7	Diffraction and Scattering	37
7.1	Bragg's Law of X-ray Diffraction	37
7.2	Powder X-ray Diffraction (PXRD)	37
7.3	Block Diagram of an X-ray Diffractometer	38
7.4	Electron and Neutron Diffraction	38
7.5	Rayleigh and Raman Scattering	38
7.6	Other Scattering Techniques	39
7.7	Applications of Diffraction and Scattering	39
7.8	Three Classical X-ray Diffraction Methods	39
7.9	Raman Effect: Quantum Picture	40
7.10	Why the Sky is Blue: A Worked Rayleigh Calculation	40
7.11	Raman Spectroscopy: Practical Applications	40

1. Principles of Spectroscopy and Selection Rules

Spectroscopy is the branch of science that studies the interaction of electromagnetic radiation with matter. The pattern of absorbed, emitted, or scattered radiation gives a fingerprint of the molecular structure, energy levels, and dynamics of the sample.

1.1 Electromagnetic Radiation and Its Quantum Nature

Electromagnetic (EM) radiation behaves both as a wave and as a stream of photons. As a wave, it is described by frequency ν , wavelength λ , and wavenumber $\bar{\nu}$. As a particle, each photon carries a discrete amount of energy $E = h\nu$.

Wave-Particle Relations

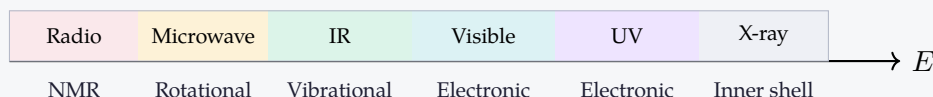
$$c = \nu\lambda, \quad \bar{\nu} = \frac{1}{\lambda} = \frac{\nu}{c}, \quad E = h\nu = \frac{hc}{\lambda} = hc\bar{\nu}$$

Symbols: $c = 3 \times 10^8 \text{ m s}^{-1}$ (speed of light); $h = 6.626 \times 10^{-34} \text{ J s}$ (Planck constant); λ in m; ν in Hz; $\bar{\nu}$ in cm^{-1} when λ is in cm.

1.2 The Electromagnetic Spectrum and Spectroscopic Regions

Different regions of the EM spectrum probe different molecular motions because each region carries an energy that matches a specific kind of energy gap.

Electromagnetic Spectrum: Energy Increases Left to Right



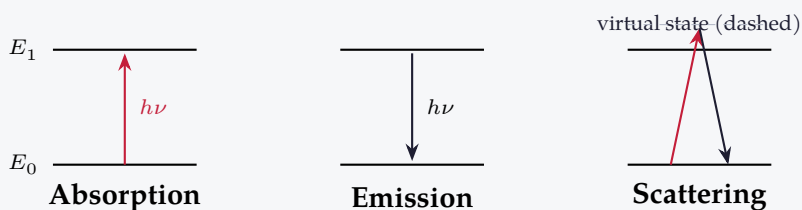
Region–energy mapping:

Region	Wavelength	Energy / quantum	Type of transition
Radio	1 m–10 km	very low (μeV)	Nuclear spin flip (NMR)
Microwave	1 mm–1 m	10^{-5} – 10^{-3} eV	Rotational (also ESR)
Infrared (IR)	700 nm–1 mm	10^{-3} –1 eV	Vibrational
Visible	400–700 nm	1.7–3.1 eV	Outer electronic
Ultraviolet (UV)	10–400 nm	3.1–124 eV	Outer electronic
X-ray	0.01–10 nm	0.1–100 keV	Inner-shell, diffraction
γ -ray	< 0.01 nm	> 100 keV	Nuclear (Mossbauer)

1.3 Absorption, Emission and Scattering

Three primary photon-matter events form the basis of every spectroscopic experiment.

Three Photon-Matter Processes



Quick comparison:

Process	What happens	Detected as	Examples
Absorption	Photon raises molecule from low to high state	Loss of intensity at ν	UV-Vis, IR
Emission	Excited molecule drops back, releasing photon	Light at characteristic ν	Fluorescence, AES
Scattering	Photon redirected (with or without energy change)	Light at angle from beam	Raman, X-ray scattering

1.4 Resonance Condition and the Bohr Frequency Rule

A photon is absorbed only when its energy matches the gap between two molecular energy levels. This is the *resonance condition*.

Bohr Frequency Condition

$$\Delta E = E_2 - E_1 = h\nu = \frac{hc}{\lambda} = hc\bar{\nu}$$

Only photons with this exact ν are absorbed. Off-resonance photons pass through unchanged. This is why spectra show *discrete lines or bands* instead of a continuum.

1.5 Selection Rules: Allowed and Forbidden Transitions

Even when the resonance condition is satisfied, a transition occurs only if the corresponding selection rule is obeyed. Selection rules come from the requirement that the transition dipole moment integral be non-zero.

Transition Probability and Dipole Moment

$$P_{1 \rightarrow 2} \propto |\mu_{12}|^2, \quad \mu_{12} = \int \psi_2^* \hat{\mu} \psi_1 d\tau$$

Allowed transition: $\mu_{12} \neq 0$ — strong band, high intensity.

Forbidden transition: $\mu_{12} = 0$ in the simple model — band absent or very weak.

Two general selection rules:

1. **Gross selection rule:** The molecule must possess a property (e.g. permanent or oscillating dipole) capable of interacting with the radiation.
2. **Specific selection rule:** The change in quantum numbers must obey definite values, e.g. $\Delta v = \pm 1$, $\Delta J = \pm 1$, $\Delta l = \pm 1$.

Selection rules across spectroscopies:

Spectroscopy	Gross rule (must have)	Specific rule
Rotational (microwave)	Permanent electric dipole moment	$\Delta J = \pm 1$
Vibrational (IR)	Vibration must change dipole moment	$\Delta v = \pm 1$ (harmonic); $\pm 1, \pm 2, \dots$ (anharmonic)
Raman	Vibration must change polarisability	$\Delta v = \pm 1$; $\Delta J = 0, \pm 2$
Electronic (UV-Vis)	Allowed orbital symmetry transition	$\Delta S = 0$ (spin); $\Delta l = \pm 1$ (Laporte)
NMR	Magnetic spin $I \neq 0$ in field B_0	$\Delta m_I = \pm 1$
ESR	Unpaired electron $S \neq 0$	$\Delta m_S = \pm 1$

Solved Example: Why N₂ Shows No IR but Shows Raman

Problem: Predict whether N₂ molecule shows IR absorption or Raman scattering.

Solution: For IR, the vibration must change the molecular dipole moment.

N₂ is homonuclear, so its dipole moment is zero throughout the stretching vibration.

⇒ IR-inactive.

For Raman, the vibration must change the polarisability of the electron cloud. The N≡N stretch does change polarisability.

⇒ Raman-active.

Mutual Exclusion Principle

For molecules with a centre of symmetry (e.g. CO₂, N₂), a vibration that is IR-active is Raman-inactive, and vice-versa. This rule is a powerful structural fingerprint and a common viva question.

1.6 Population of Energy Levels: Boltzmann Distribution

The intensity of an absorption line depends on how many molecules occupy the lower level at thermal equilibrium. The Boltzmann distribution governs this population.

Boltzmann Population Ratio

$$\frac{N_2}{N_1} = \frac{g_2}{g_1} \exp\left(-\frac{\Delta E}{k_B T}\right)$$

where g_1, g_2 are degeneracies, $k_B = 1.381 \times 10^{-23} \text{ J K}^{-1}$ and T is temperature in K.

Implication: Microwave (rotational) transitions involve very small ΔE , so excited rotational levels are well populated even at room temperature. NMR transitions involve $\Delta E \ll k_B T$,

so the population difference is tiny — that is why NMR is intrinsically a low-sensitivity technique.

1.7 Resolution and Linewidth: The Heisenberg Limit

Spectral lines are not infinitely sharp. They have natural widths set by the lifetime of the excited state through Heisenberg's uncertainty principle.

Natural Linewidth

$$\Delta E \cdot \Delta t \gtrsim \frac{\hbar}{2}, \quad \Delta \nu \approx \frac{1}{2\pi\tau}$$

A short excited-state lifetime τ produces a broad line; a long lifetime gives a sharp line.

Other broadening sources: Doppler broadening (gas phase), collisional broadening (high pressure), and instrumental broadening.

1.8 Transmittance vs Absorbance

Two scales describe how much light a sample lets through. Confusing them is the most common 2-mark error.

Transmittance and Absorbance

$$T = \frac{I}{I_0}, \quad \%T = 100 T, \quad A = -\log_{10} T = \log_{10} \frac{I_0}{I}$$

T is linear in light intensity but *non-linear* in concentration. A is linear in concentration through Beer–Lambert. Always plot A , never T , against concentration for calibration.

1.9 Einstein A and B Coefficients

Einstein analysed the equilibrium between matter and radiation by introducing three rate constants for the photon–matter exchange.

Einstein Coefficients

Stimulated absorption rate: $R_{12} = B_{12} N_1 \rho(\nu)$

Stimulated emission rate: $R_{21}^{\text{stim}} = B_{21} N_2 \rho(\nu)$

Spontaneous emission rate: $R_{21}^{\text{spon}} = A_{21} N_2$

At thermal equilibrium,

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h \nu^3}{c^3}, \quad B_{12} g_1 = B_{21} g_2$$

$\rho(\nu)$ is the radiation density at frequency ν .

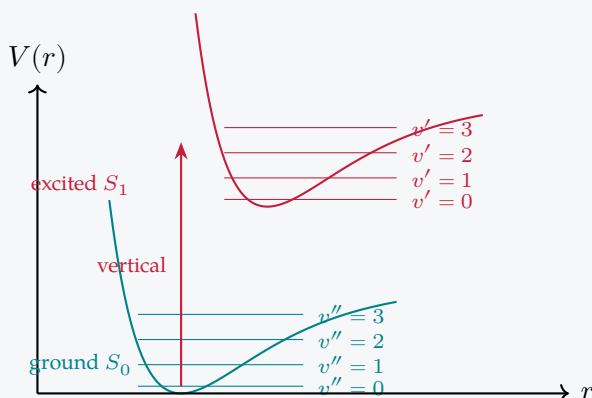
Why this matters: The ν^3 scaling shows that spontaneous emission dominates at high frequencies (visible/UV $\Rightarrow$ fluorescence) while stimulated processes dominate at low frequencies (microwave/RF $\Rightarrow$ NMR sensitivity problem). This is the theoretical basis for both lasers (population inversion makes B_{21} exceed B_{12} rates) and the weak NMR signal at room temperature.

1.10 Franck–Condon Principle

Electronic transitions are vertical on a potential-energy diagram because nuclei are too heavy to move during the 10^{-15} s duration of an electronic absorption.

Franck–Condon principle: An electronic transition is most probable when the vibrational wave functions of the two states have the largest overlap. Because the transition is vertical, the molecule lands in a vibrationally excited level of the upper electronic state.

Franck–Condon Vertical Transitions and Vibrational Progression



Consequences of the Franck–Condon principle:

- Absorption spectra of polyatomic molecules show *vibrational fine structure* (a band, not a line).
- The most intense vibronic line corresponds to the maximum-overlap (largest Franck–Condon factor) vibrational pair.
- After absorption, vibrational relaxation back to $v' = 0$ explains the Stokes shift seen in fluorescence.
- Combined with Kasha's rule (next sections), it explains why fluorescence emission spectra are wavelength-independent of the excitation λ .

Exam Points (Section 1)

- Always link region of EM spectrum to type of energy: radio–NMR, microwave–rotation, IR–vibration, UV–Vis–electronic.
- Use $E = h\nu = hc/\lambda = hc\bar{\nu}$ — examiners love unit conversion problems.
- Quote both the gross and specific selection rules; missing the gross rule is the most common cut.
- Mutual exclusion principle is a 2-mark gem; remember CO_2 .
- For NMR sensitivity question, refer to the Boltzmann ratio.
- Franck–Condon principle: “electronic transitions are vertical, nuclei don't move”.
- Einstein relation $A_{21}/B_{21} = 8\pi h\nu^3/c^3$ — explains why low-frequency NMR is

Frequently Asked / Viva Questions (contd.)

intrinsically weak.

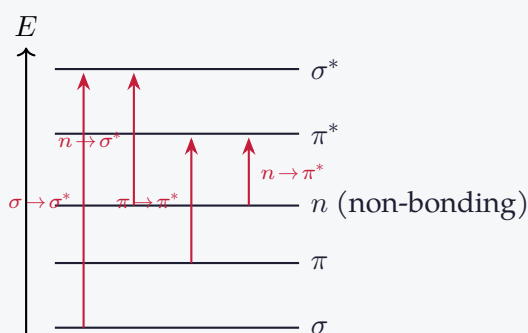
2. Electronic (UV-Visible) Spectroscopy

Electronic spectroscopy (UV-Vis) measures absorption of ultraviolet (200–400 nm) and visible (400–800 nm) light by molecules. The energy is sufficient to promote a valence electron from a filled orbital to an empty higher-energy orbital. The resulting spectrum gives information about chromophores, conjugation, and concentration.

2.1 Types of Electronic Transitions

The valence electrons of organic molecules occupy four kinds of orbitals: bonding σ and π , non-bonding n , and antibonding σ^* and π^* . The four common electronic transitions, in increasing wavelength (decreasing energy), are listed below.

Energy-Level Order of Molecular Orbitals and Possible Transitions



Comparison of the four transitions:

Transition	Compounds showing it	$\lambda_{\max}$ region	Intensity (ϵ)
$\sigma \rightarrow \sigma^*$	Saturated alkanes (CH_4 , ethane)	< 150 nm (vacuum UV)	High but inaccessible
$n \rightarrow \sigma^*$	H_2O , CH_3OH , alkyl halides, amines	150–250 nm	Medium (10^2 – 10^3)
$\pi \rightarrow \pi^*$	Alkenes, alkynes, aromatics, carbonyls	170–250 nm	High (10^3 – 10^4)
$n \rightarrow \pi^*$	Aldehydes, ketones, $> \text{C}=\text{O}$, $> \text{C}=\text{N}$	270–300 nm	Low (10 – 10^2)

2.2 Beer–Lambert Law: The Quantitative Backbone

UV-Vis is a quantitative tool because the absorbance is linearly related to concentration through the Beer–Lambert law.

Beer–Lambert Law

$$A = \log_{10} \frac{I_0}{I} = \varepsilon c l$$

Symbols: A = absorbance (dimensionless); I_0 = incident intensity; I = transmitted intensity; ε = molar absorptivity ($\text{L mol}^{-1} \text{cm}^{-1}$); c = concentration (mol L^{-1}); l = path length (cm).

Also useful:

$$T = \frac{I}{I_0}, \quad A = -\log_{10} T = 2 - \log_{10}(\%T)$$

Limitations of Beer–Lambert law:

Cause of deviation	Reason
High concentration ($> 10^{-2} \text{ M}$)	Solute molecules interact, ε is no longer constant
Polychromatic light	Different wavelengths have different ε , average becomes non-linear
Stray radiation	Light reaching detector by leakage flattens absorbance at high A
Chemical equilibria	Dimerisation, ionisation, fluorescence change effective absorbing species

Solved Example: Absorbance to Concentration

Problem: A solution of an organic dye gives $A = 0.92$ in a 1.0 cm cell at 510 nm, where $\varepsilon = 2300 \text{ L mol}^{-1} \text{cm}^{-1}$. Find the concentration.

Solution: From $A = \varepsilon c l$,

$$c = \frac{A}{\varepsilon l} = \frac{0.92}{2300 \times 1.0} = 4.0 \times 10^{-4} \text{ mol L}^{-1}$$

Exam tip: Always state whether you used $\log_{10}$ or $\ln$. The IUPAC form uses $\log_{10}$.

2.3 Chromophores and Auxochromes

Functional groups that determine where a molecule absorbs are called chromophores, while certain saturated groups that shift the band of a chromophore are called auxochromes.

Chromophore: Unsaturated functional group responsible for absorption (e.g. $\text{C}=\text{C}$, $\text{C}=\text{O}$, $\text{N}=\text{N}$, NO_2).

Auxochrome: Saturated group with non-bonding electrons (e.g. $-\text{OH}$, $-\text{NH}_2$, $-\text{OR}$, $-\text{X}$) that, when attached to a chromophore, shifts λ_{max} to longer wavelength and increases intensity.

Standard wavelength-shift terminology:

Term	Effect on $\lambda_{\max}$	Cause
Bathochromic (red) shift	$\lambda_{\max}$ <i>increases</i>	Conjugation, auxochrome, polar solvent (for $\pi \rightarrow \pi^*$)
Hypsochromic (blue) shift	$\lambda_{\max}$ <i>decreases</i>	Removal of conjugation, polar solvent (for $n \rightarrow \pi^*$)
Hyperchromic effect	ϵ <i>increases</i>	Increased symmetry/conjugation
Hypochromic effect	ϵ <i>decreases</i>	Distortion of chromophore

2.4 Effect of Conjugation

Each additional conjugated C=C raises the highest occupied π MO and lowers the lowest empty π^* MO. The HOMO–LUMO gap shrinks, so $\lambda_{\max}$ moves to longer wavelength.

Compound	Conjugated double bonds	$\lambda_{\max}$ (nm)
Ethylene	1	165
1,3-Butadiene	2	217
1,3,5-Hexatriene	3	258
β-Carotene	11	470 (orange colour)

2.5 Solvent Effects on Electronic Spectra

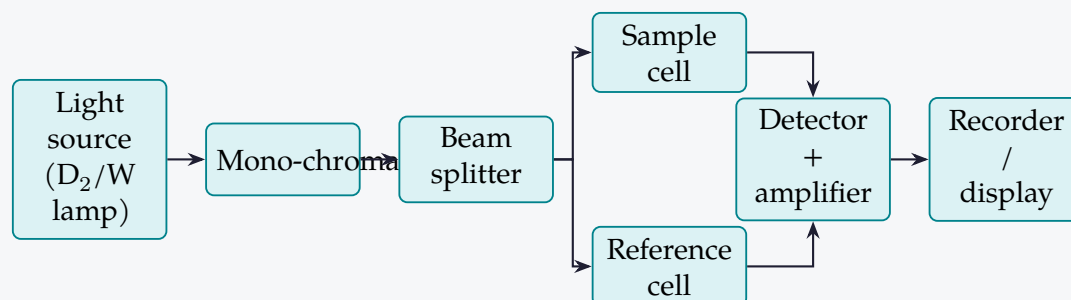
The solvent stabilises the ground and excited states differently, so the absorption band shifts.

Transition	Polar protic solvent effect	Reason
$\pi \rightarrow \pi^*$	Bathochromic (red shift)	Excited state (π^*) more polar, better stabilised
$n \rightarrow \pi^*$	Hypsochromic (blue shift)	Hydrogen bonds with lone pair lower the ground state more

2.6 Block Diagram of a UV-Visible Spectrophotometer

A double-beam spectrophotometer compares the sample beam with a reference beam in real time, eliminating fluctuations of the lamp.

Double-Beam UV-Vis Spectrophotometer



Common light sources and their range:

- Deuterium (D₂) lamp — UV region, 190–400 nm.
- Tungsten–halogen lamp — visible region, 350–2500 nm.
- Xenon flash lamp — both UV and visible.

2.7 Applications of UV-Visible Spectroscopy

- Quantitative estimation of metal ions (Fe³⁺ as thiocyanate, etc.).
- Determination of molar absorptivity and concentration in pharmaceuticals.
- Detection of conjugated systems and aromatic rings.
- Reaction kinetics: change in A with time gives the rate.
- Identification of dyes and impurity profiling.

2.8 Woodward–Fieser Rules

For dienes and α, β -unsaturated carbonyls, $\lambda_{\max}$ can be predicted to within ± 5 nm by adding empirical increments to a base value. This is a MAKAUT favourite for 5- and 10-mark questions.

Woodward–Fieser Rule for Dienes

$$\lambda_{\max} = \lambda_{\text{base}} + \sum (\text{increments})$$

Base values (in nm):

Acyclic / heteroannular diene: **215** Homoannular diene: **253**

Common increments:

- Each extra conjugated C=C: +30
- Alkyl group / ring residue: +5
- Exocyclic double bond: +5
- –OR: +6, –OAc: 0, –Cl/ – Br: +5, –NR₂: +60

Solved Example: Predicting $\lambda_{\max}$ of 2,4-Dimethyl-1,3-pentadiene**Structure:** acyclic diene with two methyl substituents and one ring residue.**Calculation:**

- Base value (acyclic diene)215
- Three alkyl groups $\times + 5$ +15

$$\lambda_{\max}^{\text{pred}} = 215 + 15 = 230 \text{ nm}$$

Reported: $\lambda_{\max} = 232 \text{ nm}$ — within the $\pm 5 \text{ nm}$ rule.

Woodward–Fieser rule for enones: The rule extends to α, β -unsaturated ketones with base value 215 nm (acyclic / six-ring) and base value 202 nm (five-ring). Each α -substituent adds 10 nm, β -substituent adds 12 nm, and so on. Beyond the engineering syllabus the principle is the same — pile up empirical increments on a base.

2.9 Isosbestic Point

When two species *A* and *B* inter-convert (e.g. acid–base equilibrium or a kinetic transformation), their UV-Vis spectra cross at one or more wavelengths where the absorbance does *not* change with composition.

Isosbestic point: A wavelength at which two absorbing species have the same molar absorptivity. The presence of a sharp isosbestic point in a series of recorded spectra is direct evidence of a clean two-component equilibrium without any third absorbing intermediate.

Why exam-relevant: In titrations, kinetic studies, and pH-dependent measurements the isosbestic point lets you measure total concentration independent of the equilibrium position.

2.10 Single-Beam vs Double-Beam Instruments

Feature	Single-beam	Double-beam
Light path	One path: source $\rightarrow$ sample $\rightarrow$ detector	Two paths simultaneously: sample and reference
Reference correction	Manual (run blank, then sample)	Automatic, real-time
Lamp drift	Affects reading	Cancelled out
Cost	Cheaper, simpler	More expensive but more accurate
Use case	Routine fixed- λ measurements	Spectral scanning, kinetics

2.11 Sample Cell (Cuvette) Materials

Choosing the wrong cuvette is the most common practical mistake — quartz absorbs nothing in UV, glass cuts off below 340 nm.

Material	Useful range	Notes
Quartz / fused silica	190–2500 nm	Mandatory for UV; expensive
Borosilicate glass	340–2500 nm	Visible/NIR only
Polystyrene (disposable)	340–800 nm	Single-use, cheap, visible only
CaF ₂ / BaF ₂	150 nm–mid-IR	Vacuum-UV and IR work

Exam Points (Section 2)

- State all four transitions with energy order: $n \rightarrow \pi^* < \pi \rightarrow \pi^* < n \rightarrow \sigma^* < \sigma \rightarrow \sigma^*$.
- For Beer–Lambert numericals, watch units of c and l .
- Always link conjugation with bathochromic shift (β -carotene, lycopene).
- Mention deuterium lamp for UV and tungsten lamp for visible.
- Use the Woodward–Fieser increment formula for enone / diene predictions.
- Quartz cuvette below 340 nm, glass above 340 nm.
- Sharp isosbestic point = clean two-component equilibrium.

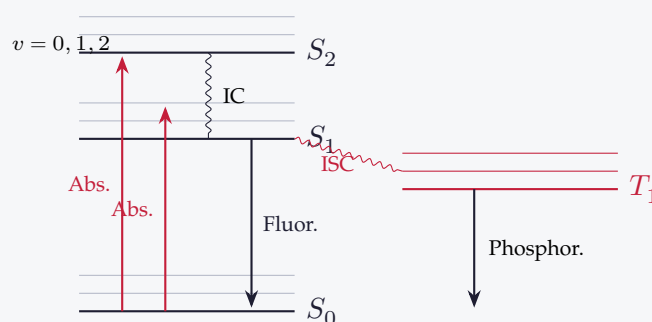
3. Fluorescence and Its Applications in Medicine

Fluorescence is the spontaneous emission of light by a molecule that has been excited to a higher singlet electronic state, when it returns to its ground state *without changing spin multiplicity*. The emission lifetime is short (10^{-9} – 10^{-7} s), and the emission stops as soon as the exciting source is turned off.

3.1 Singlet, Triplet and the Jablonski Diagram

A molecule can have its electron pair with antiparallel spins (singlet, S) or parallel spins (triplet, T). All photophysical processes can be summarised in a Jablonski diagram.

Jablonski Energy-Level Diagram



Key processes shown in the Jablonski diagram:

- **Absorption:** $S_0 \rightarrow S_1$ or $S_0 \rightarrow S_2$ in 10^{-15} s.
- **Vibrational relaxation (VR):** loss of excess vibrational energy in 10^{-12} s.
- **Internal conversion (IC):** non-radiative $S_2 \rightarrow S_1$, same multiplicity.
- **Fluorescence (F):** radiative $S_1 \rightarrow S_0$ in 10^{-9} – 10^{-7} s.
- **Inter-system crossing (ISC):** $S_1 \rightarrow T_1$, spin-forbidden but possible.
- **Phosphorescence (P):** radiative $T_1 \rightarrow S_0$ in 10^{-3} – 10 s.

3.2 Fluorescence vs Phosphorescence

Property	Fluorescence	Phosphorescence
Initial state	Singlet excited (S_1)	Triplet excited (T_1)
Spin change	None (allowed)	Yes (formally forbidden)
Lifetime	10^{-9} – 10^{-7} s	10^{-3} – 10 s
After source off	Stops immediately	Persists (afterglow)
Wavelength	Slightly higher than absorption	Even higher than fluorescence
Common at	Room temperature, fluid solution	Low temperature, rigid medium

3.3 Stokes Shift and Mirror-Image Rule

Stokes shift is the difference between the absorption maximum and the emission maximum, $\Delta\lambda = \lambda_{\text{em}} - \lambda_{\text{abs}}$. It arises because the molecule loses vibrational energy before emitting.

Mirror-image rule: For most rigid fluorophores the emission spectrum is approximately a mirror image of the absorption spectrum on the wavelength axis.

3.4 Quantum Yield and Quenching

Quantum Yield of Fluorescence

$$\Phi_F = \frac{\text{number of photons emitted}}{\text{number of photons absorbed}} = \frac{k_F}{k_F + k_{NR} + k_{ISC} + k_Q[Q]}$$

k_F is the radiative rate; k_{NR} , k_{ISC} , k_Q are non-radiative, inter-system crossing and quenching rates respectively. $0 \leq \Phi_F \leq 1$.

Stern–Volmer equation for collisional quenching:

$$\frac{F_0}{F} = 1 + K_{SV}[Q]$$

where F_0 is fluorescence intensity without quencher, F is intensity in presence of quencher concentration $[Q]$, and K_{SV} is the Stern–Volmer constant.

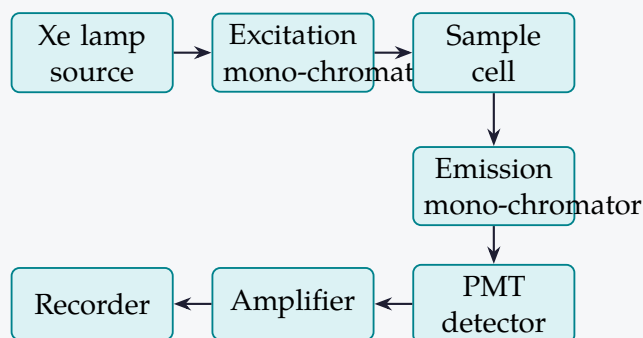
3.5 Factors Influencing Fluorescence Intensity

Factor	Effect
Rigidity of molecule	Rigid planar systems (fluorescein, rhodamine) fluoresce strongly because non-radiative pathways are suppressed.
Conjugation / aromaticity	Extended π -system shifts emission to longer λ and increases Φ_F .
Substituents	Electron-donating ($-\text{NH}_2$, $-\text{OH}$) enhance; heavy atoms (Br, I) and NO_2 promote ISC and quench.
Solvent	Polar solvent stabilises the more polar excited state $\Rightarrow$ red-shifted, sometimes weaker emission.
Temperature	Higher T increases collisional non-radiative loss $\Rightarrow$ weaker fluorescence.
Dissolved O_2	Triplet O_2 quenches both fluorescence and phosphorescence.
pH	Protonation/deprotonation of acidic chromophores changes Φ_F and λ_{em} .

3.6 Spectrofluorimeter (Block Diagram)

A fluorimeter detects light at 90° to the excitation beam to avoid interference from transmitted light.

Block Diagram of a Spectrofluorimeter



3.7 Applications in Medicine and Biology

Because fluorescence detection can sense single molecules, it has revolutionised diagnostics and therapy. The main medical applications are listed below.

Application	How fluorescence is used
Clinical assays (immunoassay)	Antibodies tagged with fluorescein/rhodamine bind specific antigens; concentration is read from fluorescence intensity (e.g. FIA, FACS).
DNA / RNA detection	Intercalating dyes (ethidium bromide, SYBR Green) and fluorescent probes used in gel imaging and <i>quantitative</i> real-time PCR.
Fluorescence microscopy	GFP-tagged proteins visualised inside living cells; reveals localisation and dynamics.
Fluorescent angiography	Sodium fluorescein injected intravenously; ophthalmologist captures retinal vasculature images.
Cancer detection / surgery	5-Aminolaevulinic acid (5-ALA) accumulates as fluorescent protoporphyrin IX in glioma cells; surgeon visualises tumour boundary in blue light.
Photodynamic therapy (PDT)	Light excites a porphyrin photosensitiser; via ISC it generates singlet oxygen which destroys tumour cells.
Pulse oximetry-related	Near-IR fluorescent probes monitor oxygenation; complementary to absorption oximetry.
Drug binding studies	Tryptophan in proteins (intrinsic fluorophore) reports drug-binding through quenching of its 340 nm band.
Biosensors	Fluorescence quenching of conjugated polymers used in glucose, Ca^{2+} and pH sensing.

Solved Example: Stern–Volmer Plot Reading

Problem: A fluorescence quenching experiment gives the following data:

$$[Q] = 0, 1, 2 \text{ mM}; \quad F = 100, 67, 50 \text{ (arb.)}$$

Find K_{SV} .

Solution: Compute F_0/F for each $[Q]$:

$$F_0/F = 1.00, 1.49, 2.00 \Rightarrow \text{slope} \approx 0.5 \text{ mM}^{-1}$$

$$K_{SV} = 500 \text{ M}^{-1}$$

A linear plot confirms purely collisional (dynamic) quenching.

Why Heavy Atoms and Oxygen Quench Fluorescence

Heavy atoms (Br, I) and dissolved O_2 promote inter-system crossing $S_1 \rightarrow T_1$. Once a molecule is in the triplet state, fluorescence is no longer possible. This “heavy-atom effect” is the basis of phosphorescent compounds, and the reason why deoxygenation is needed for high-quantum-yield emission measurements.

3.8 Kasha's Rule

Kasha's rule: Emission (fluorescence or phosphorescence) almost always occurs from the *lowest* excited state of a given multiplicity, regardless of which higher state was initially populated by the absorbed photon.

Why it matters:

- Internal conversion $S_n \rightarrow S_1$ is much faster than $S_1 \rightarrow S_0$ fluorescence.
- The fluorescence wavelength is therefore independent of the excitation wavelength.
- This is why an excitation spectrum (intensity vs λ_{exc}) reproduces the absorption spectrum exactly.
- Combined with the Franck–Condon principle (Section 1.7), it gives the mirror-image rule.

3.9 Lifetime, Quantum Yield, and Radiative Rate

The radiative lifetime τ_F and quantum yield Φ_F are not independent — they jointly fix the natural radiative rate k_F .

Lifetime–Quantum Yield Connection

$$\tau_F = \frac{1}{k_F + k_{NR} + k_{ISC}}, \quad \Phi_F = k_F \tau_F$$

The natural (radiative) lifetime is $\tau_0 = 1/k_F = \tau_F/\Phi_F$.

A short measured τ_F together with low Φ_F indicates strong non-radiative deactivation.

Diagnostic use: Time-resolved fluorimetry measures τ_F directly. Combined with the steady-state Φ_F , you can separate radiative (k_F) from non-radiative ($k_{NR} + k_{ISC}$) contributions and identify the dominant decay pathway.

3.10 Förster Resonance Energy Transfer (FRET)

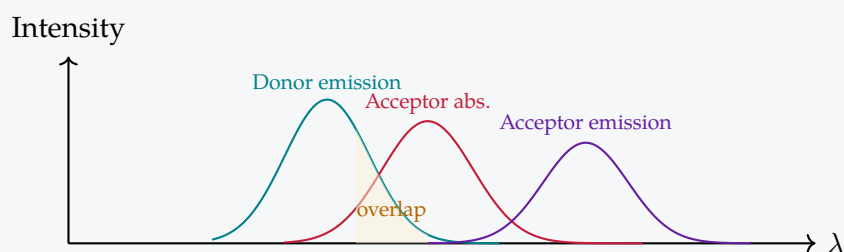
FRET is non-radiative dipole–dipole energy transfer from an excited donor (D^*) to a nearby ground-state acceptor (A). The efficiency depends very strongly on the donor–acceptor distance, making FRET a “spectroscopic ruler” on the 1–10 nm scale.

FRET Efficiency

$$E = \frac{R_0^6}{R_0^6 + r^6} = 1 - \frac{\tau_{DA}}{\tau_D}$$

r = donor–acceptor distance; R_0 = Förster radius (typically 1–10 nm, the distance at which $E = 50\%$); τ_D and τ_{DA} are donor lifetimes alone and in presence of acceptor. Conditions: spectral overlap of donor emission and acceptor absorption; favourable orientation; r within $\sim 2R_0$.

FRET: Donor and Acceptor Spectral Overlap



Medical and biological applications of FRET:

- Protein conformational changes (label two residues with D and A).
- DNA hybridisation assays and SNP detection.
- Real-time PCR with TaqMan probes.
- Single-molecule FRET to track folding/binding kinetics.
- Fluorescent biosensors for Ca^{2+} , Zn^{2+} , ATP, kinase activity.

Exam Points (Section 3)

- Always draw the Jablonski diagram first; mark S_0 , S_1 , T_1 , IC, ISC, F, P.
- Stokes shift = “vibrational relaxation before emission”.
- Compare F and P using a 4-row table.
- Medicine answer must include *at least three* applications: fluorescein angiography, PDT, FIA.
- State Kasha’s rule and link it to the wavelength-independence of emission.
- FRET efficiency $E = R_0^6/(R_0^6 + r^6)$ — the r^{-6} distance dependence is the standard 5-mark question.

4. Vibrational and Rotational Spectroscopy of Diatomic Molecules

A diatomic molecule rotates about its centre of mass and vibrates along its bond. Photon absorption in the microwave region excites pure rotation; absorption in the infrared region excites vibration (often together with simultaneous rotational changes). Together these tell us bond length, bond strength, force constant, and moment of inertia.

4.1 Rotational Spectroscopy: The Rigid-Rotor Model

The molecule is modelled as two masses m_1 and m_2 joined by a rigid rod of length r (the bond length). The reduced mass is

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

and the moment of inertia about the centre of mass is

$$I = \mu r^2$$

Rotational Energy Levels (Rigid Rotor)

$$E_J = \frac{\hbar^2}{2I} J(J+1) \quad \text{or} \quad \tilde{E}_J = \bar{B} J(J+1) \text{ cm}^{-1}$$

where $J = 0, 1, 2, \dots$ is the rotational quantum number and the rotational constant is

$$\bar{B} = \frac{h}{8\pi^2 c I} \text{ cm}^{-1}$$

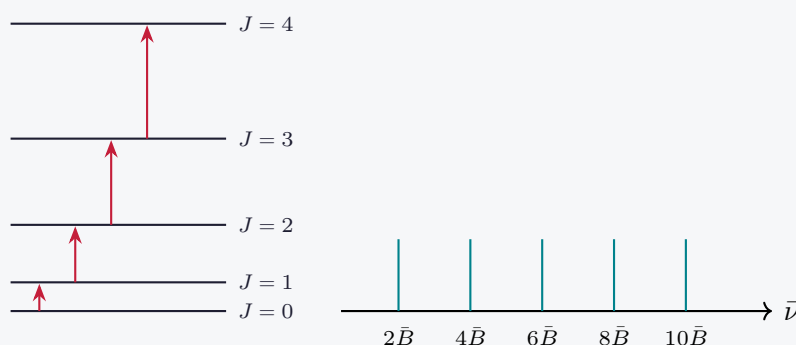
Selection rule: $\Delta J = \pm 1$, and the molecule must possess a permanent dipole moment.

Spacing of allowed transitions: For absorption $J \rightarrow J+1$,

$$\Delta \tilde{E} = \tilde{E}_{J+1} - \tilde{E}_J = 2\bar{B}(J+1)$$

so successive lines are separated by a constant $2\bar{B}$. From this spacing one can compute I , then r .

Rotational Energy Levels and Spectrum



Solved Example: Bond Length of HCl from Rotational Spectrum

Problem: For HCl, $\bar{B} = 10.59 \text{ cm}^{-1}$. Calculate the H-Cl bond length. Take $\mu_{\text{HCl}} = 1.628 \times 10^{-27} \text{ kg}$.

Solution: From $\bar{B} = h/(8\pi^2 cI)$,

$$I = \frac{h}{8\pi^2 c \bar{B}} = \frac{6.626 \times 10^{-34}}{8\pi^2 \times 3 \times 10^{10} \times 10.59} = 2.64 \times 10^{-47} \text{ kg m}^2$$

Using $I = \mu r^2$,

$$r = \sqrt{\frac{I}{\mu}} = \sqrt{\frac{2.64 \times 10^{-47}}{1.628 \times 10^{-27}}} = 1.27 \times 10^{-10} \text{ m} = 127 \text{ pm}$$

Result: The H-Cl bond length is $\approx 127 \text{ pm}$.

4.2 Vibrational Spectroscopy: Simple Harmonic Oscillator Model

For small displacements the bond can be modelled as a Hookean spring with force constant k . Its quantised energies are

$$E_v = \left(v + \frac{1}{2}\right) h\nu_0, \quad v = 0, 1, 2, \dots, \quad \nu_0 = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$

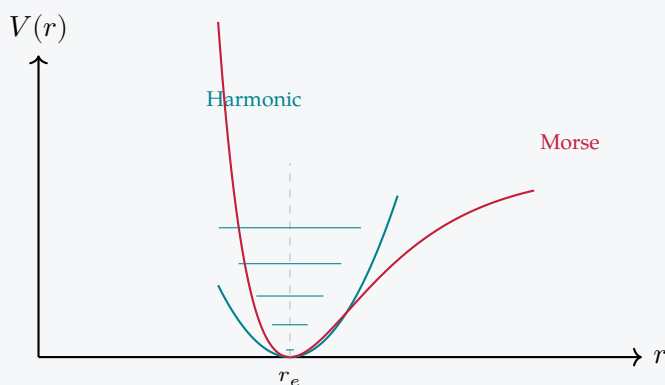
The fundamental absorption is the $v=0 \rightarrow v=1$ transition.

Vibrational Frequency and Wavenumber

$$\bar{\nu}_0 = \frac{1}{2\pi c} \sqrt{\frac{k}{\mu}} \text{ cm}^{-1}$$

Selection rule: $\Delta v = \pm 1$ for harmonic oscillator (gross rule: vibration must change the dipole moment).

Zero-point energy: $E_0 = \frac{1}{2} h\nu_0 \neq 0$, even at 0 K.

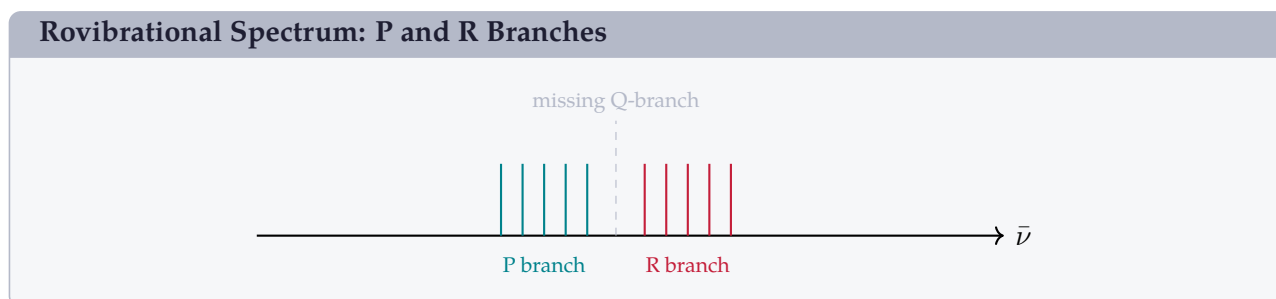
Harmonic Oscillator and Anharmonic (Morse) Potential

Anharmonicity (Morse) corrections:

Feature	Effect
Levels not equally spaced	Spacing decreases as v increases.
Selection rule modified	Overtone $\Delta v = \pm 2, \pm 3, \dots$ become weakly allowed.
Bond dissociation possible	At sufficient v , vibration breaks the bond.
Energy expression	$\tilde{E}_v = \bar{\nu}_e(v + \frac{1}{2}) - \bar{\nu}_e x_e(v + \frac{1}{2})^2$

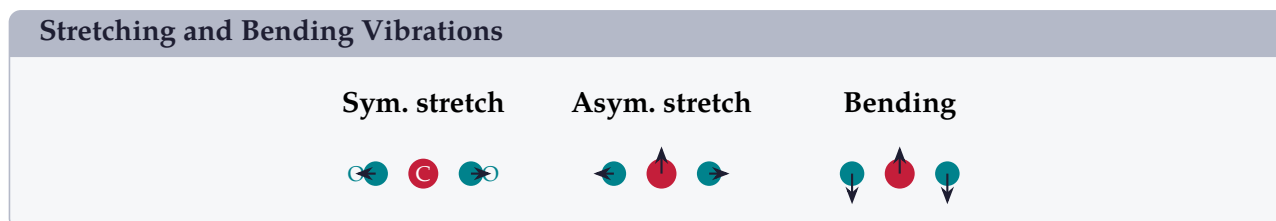
4.3 Vibration–Rotation (Rovibrational) Spectra of Diatomics

A vibrational transition is usually accompanied by a change in J . The fundamental band of HCl, for example, shows a P-branch ($\Delta J = -1$) and an R-branch ($\Delta J = +1$) but *no* central Q-branch ($\Delta J = 0$).



4.4 Modes of Vibration in Polyatomic Molecules

A non-linear molecule with N atoms has $3N - 6$ vibrational modes; a linear one has $3N - 5$ modes. Each mode is either stretching (along the bond) or bending (across the bond).



Number of modes for common molecules:

Molecule	Geometry	$3N - 5$ or $3N - 6$	Modes
CO ₂	Linear	$3(3) - 5$	4 (sym str, asym str, two bends)
H ₂ O	Bent	$3(3) - 6$	3 (sym str, asym str, scissor)
NH ₃	Pyramidal	$3(4) - 6$	6
CH ₄	Tetrahedral	$3(5) - 6$	9

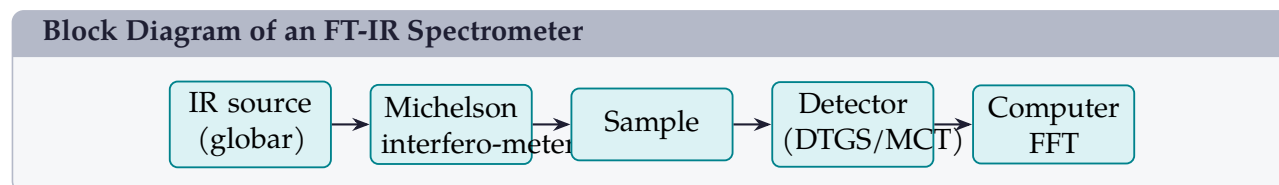
4.5 Characteristic IR Group Frequencies

A practical IR fingerprint table is essential for compound identification.

Group	Wavenumber (cm^{-1})	Intensity / nature
O–H (alcohol)	3200–3550 (broad)	Strong, broad due to H-bond
N–H (amine)	3300–3500	Medium, sharp
C–H (sp^3)	2850–2960	Strong
C–H (sp^2 , aromatic)	3000–3100	Medium
C $\equiv$ N	2210–2260	Medium
C $\equiv$ C	2100–2260	Weak
C=O (ester, ketone)	1680–1750	Very strong
C=C	1620–1680	Variable
N–O (NO_2)	1300–1550 (two bands)	Strong
C–O	1000–1300	Strong

4.6 Block Diagram of an FT-IR Spectrometer

Modern IR instruments use a Michelson interferometer and Fourier transform of the interferogram to obtain the spectrum.



4.7 Applications of IR and Microwave Spectroscopy

- Identification of functional groups (most common organic-chem application).
- Distinguishing isomers (e.g. acetone vs propanal by C=O stretch position).
- Hydrogen bonding studies through O–H shift.
- Quantitative analysis using calibrated absorbance.
- Microwave spectroscopy gives bond lengths and bond angles to better than 0.1 pm.
- Pollution monitoring: IR detects CO, CO₂, NO_x, SO₂ in stack gas.
- Polymer/coating analysis (ATR-IR).

4.8 Centrifugal Distortion of the Rigid Rotor

At higher rotational levels the bond stretches slightly under centrifugal force, increasing the moment of inertia and reducing the spacing between successive lines. The corrected energy

expression is

$$\tilde{E}_J = \bar{B} J(J+1) - \bar{D} J^2(J+1)^2$$

where $\bar{D}$ is the centrifugal-distortion constant ($\bar{D} \ll \bar{B}$). This makes high- J lines slightly closer than the rigid-rotor prediction of $2\bar{B}$ spacing.

4.9 Isotope Effect on Vibrational and Rotational Spectra

Replacing H by D doubles the reduced mass without changing the force constant. This is a standard MAKAUT viva question and a standard 5-mark numerical.

Effect of Isotopic Substitution

Since $\bar{\nu}_0 = (1/2\pi c)\sqrt{k/\mu}$ and k is unchanged,

$$\frac{\bar{\nu}_D}{\bar{\nu}_H} = \sqrt{\frac{\mu_H}{\mu_D}} \approx \frac{1}{\sqrt{2}} \approx 0.71$$

For rotational spectra, $\bar{B} \propto 1/\mu r^2$, so heavier isotopes give smaller $\bar{B}$ and tighter line spacing.

Solved Example: H–Cl vs D–Cl Vibrational Frequency

For HCl, $\bar{\nu}_0 \approx 2989 \text{ cm}^{-1}$. Predict $\bar{\nu}_0$ for DCl.

The reduced masses are $\mu_{\text{HCl}} = 0.972 \text{ u}$ and $\mu_{\text{DCl}} = 1.904 \text{ u}$.

$$\bar{\nu}_{\text{DCl}} = \bar{\nu}_{\text{HCl}} \sqrt{\frac{\mu_{\text{HCl}}}{\mu_{\text{DCl}}}} = 2989 \sqrt{\frac{0.972}{1.904}} = 2989 \times 0.715 = 2137 \text{ cm}^{-1}$$

Observed value: 2145 cm^{-1} — excellent agreement.

4.10 Fingerprint Region vs Functional-Group Region

The IR spectrum is conventionally divided into two regions, and they are used differently in identification.

Region	Range (cm^{-1})	Use
Functional-group region	4000–1500	Quick identification of common groups (O–H, N–H, C–H, C=O, C≡N, etc.).
Fingerprint region	1500–400	Highly specific to the whole molecule; used to confirm identity by comparison with a reference spectrum. Do not assign every peak; match the overall pattern.

4.11 Sample Preparation for IR Spectroscopy

The right method depends on the physical state of the sample.

Sample	Method	Notes
Solid (crystalline)	KBr pellet	1 % sample in dry KBr, pressed at 10 tonnes; KBr is IR-transparent above 400 cm^{-1} .
Solid (amorphous)	Nujol mull	Mineral oil paste between two NaCl plates. Nujol bands at $\sim 2900, 1460, 1380\text{ cm}^{-1}$ — ignore them.
Liquid (neat)	Capillary film	Single drop between two NaCl windows.
Liquid (dilute)	Solution cell	CCl_4 , CHCl_3 , or CS_2 — all transparent in different ranges.
Gas	Long-path gas cell	5–10 cm path, NaCl/KBr windows.
Surface / film	ATR (attenuated total reflectance)	No prep needed; press sample on Ge/diamond crystal.

4.12 IR vs Raman: Side-by-Side Comparison

Feature	IR	Raman
Photon process	Absorption	Inelastic scattering
Gross selection rule	Vibration must change <i>dipole</i>	Vibration must change <i>polarisability</i>
Strongest for	Polar bonds (O–H, C=O)	Symmetric, non-polar bonds (C=C, S–S)
Water as solvent	Bad (water absorbs strongly)	Excellent (water is a weak Raman scatterer)
Sample prep	KBr pellet, mull, ATR	Almost none; can be in glass capillary
Source	IR global	Visible / NIR laser
Sensitivity	High	Low (only $\sim 10^{-7}$ of incident photons scattered)

Exam Points (Section 4)

- Quote both the gross rule (dipole change) and specific rule ($\Delta v = \pm 1$) for IR.
- For numericals: $\bar{B} = h/(8\pi^2 cI)$ and $\bar{\nu}_0 = (1/2\pi c)\sqrt{k/\mu}$.
- Mention zero-point energy as a quantum effect.
- For polyatomic, give $3N - 6 / 3N - 5$ rule with $\text{H}_2\text{O}/\text{CO}_2$ example.
- Always cite C=O ($\sim 1700\text{ cm}^{-1}$) and O–H ($\sim 3400\text{ cm}^{-1}$) bands.
- Isotope shift $\bar{\nu}_D/\bar{\nu}_H \approx 1/\sqrt{2}$ for X–H vs X–D — standard 5-mark numerical.

Frequently Asked / Viva Questions (contd.)

- Sample prep choice: KBr pellet for solids, ATR for everything else.
- Fingerprint region $< 1500\text{ cm}^{-1}$ confirms identity, functional-group region $> 1500\text{ cm}^{-1}$ identifies groups.

5. Nuclear Magnetic Resonance (NMR) and Magnetic Resonance Imaging (MRI)

Nuclear Magnetic Resonance (NMR) is the absorption of radio-frequency radiation by nuclei placed in a strong magnetic field. Only nuclei with non-zero spin quantum number $I \neq 0$ are NMR-active (e.g. ^1H , ^{13}C , ^{19}F , ^{31}P). The chemical environment of each nucleus modifies the resonance frequency and gives a structural fingerprint of the molecule.

5.1 Nuclear Spin and Magnetic Moment

A nucleus with spin I has a magnetic moment

$$\vec{\mu} = \gamma \vec{I} \hbar$$

where γ is the magnetogyric ratio. In an external field $\vec{B}_0$, the spin can take $2I+1$ orientations of energy

$$E_m = -\gamma m_I \hbar B_0, \quad m_I = I, I-1, \dots, -I$$

For a ^1H nucleus ($I = 1/2$) only two states exist: $m_I = +1/2$ (lower, aligned with B_0) and $m_I = -1/2$ (higher, opposed).

Larmor (Resonance) Condition

$$\Delta E = \gamma \hbar B_0 = h \nu_L \Rightarrow \nu_L = \frac{\gamma B_0}{2\pi}$$

ν_L is called the Larmor frequency. **Selection rule:** $\Delta m_I = \pm 1$.
For ^1H in $B_0 = 11.74\text{ T}$, $\nu_L = 500\text{ MHz}$.

5.2 Chemical Shift

The local electron cloud shields each nucleus, so the actual field experienced is $B_{\text{eff}} = B_0(1 - \sigma)$, where σ is the shielding constant. Different chemical environments have different σ , leading to slightly different resonance frequencies. To make the values instrument-independent, the chemical shift is referenced to TMS (tetramethylsilane).

Chemical Shift

$$\delta \text{ (ppm)} = \frac{\nu_{\text{sample}} - \nu_{\text{TMS}}}{\nu_{\text{spectrometer}}} \times 10^6$$

Higher δ = deshielded = downfield. Lower δ = shielded = upfield. TMS is set at $\delta = 0$ ppm.

Typical ^1H chemical shift ranges:

Type of proton	δ (ppm)	Comment
TMS	0.0	Reference standard
Alkyl C–H	0.5–2.0	Most shielded
Allylic / α -carbonyl	2.0–3.0	Slight deshielding
C–O–H, C–O–CH	3.0–4.5	Deshielded by O
Vinyl (C=CH)	4.5–6.5	π -electron shielding
Aromatic	6.5–8.5	Ring current deshielding
Aldehyde (CHO)	9.0–10.0	Strong deshielding by C=O
Carboxylic acid OH	10–13 (broad)	H-bonded, broad

5.3 Spin–Spin Coupling and Multiplicity

Adjacent magnetic nuclei split each other's signals. The number of equivalent neighbours n produces $n + 1$ lines in a binomial intensity ratio (Pascal's triangle).

$n + 1$ Rule

$$\text{multiplicity} = n + 1$$

where n is the number of *equivalent* neighbouring (vicinal) protons. The coupling constant J is measured in Hz and is independent of B_0 .

Common multiplets:

n neighbours	Pattern	Intensity ratio	Example
0	Singlet	1	$(\text{CH}_3)_4\text{Si}$
1	Doublet	1:1	$\text{CHCl}_2\text{--CH}$ (coupled to 1H)
2	Triplet	1:2:1	$\text{CH}_3\text{--CH}_2\text{--X}$
3	Quartet	1:3:3:1	$\text{X--CH}_2\text{--CH}_3$
6	Septet	1:6:15:20:15:6:1	Isopropyl CH

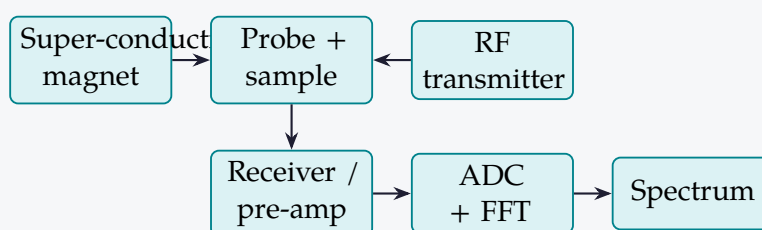
Solved Example: NMR of Ethanol ($\text{CH}_3\text{CH}_2\text{OH}$)**Three sets of equivalent protons:**

- CH_3 : $\delta \approx 1.2$ ppm, **triplet** (coupled to 2 CH_2 neighbours).
- CH_2 : $\delta \approx 3.7$ ppm, **quartet** (coupled to 3 CH_3 neighbours).
- OH : $\delta \approx 2.6$ ppm, broad **singlet** (rapid exchange averages coupling).

Integration ratio: 3 : 2 : 1, exactly matching the proton counts.

5.4 Block Diagram of an NMR Spectrometer

Modern instruments are pulsed Fourier-transform spectrometers (FT-NMR), which excite all nuclei simultaneously with a strong RF pulse and detect the free-induction decay (FID).

Block Diagram of FT-NMR**5.5 Applications of NMR**

- Structural elucidation of organic compounds (number of H environments, connectivity, stereochemistry).
- Molecular dynamics: relaxation times T_1 and T_2 probe motion.
- Quantitative NMR (qNMR) for drug purity certification.
- Solid-state NMR (MAS) for polymers, batteries, and materials.
- Protein structure determination via 2D NOESY/HSQC.

5.6 Magnetic Resonance Imaging (MRI)

MRI applies the principles of NMR to image the inside of the human body without ionising radiation.

MRI produces tomographic images by spatially encoding the NMR signal of water protons in the body using *magnetic field gradients*, then reconstructing the image by Fourier transform.

Why protons? The body contains $\sim 60\text{--}70\%$ water, providing an abundant ^1H signal. Different tissues differ in (i) proton density, (ii) longitudinal relaxation time T_1 , and (iii) transverse relaxation time T_2 — the key contrast parameters.

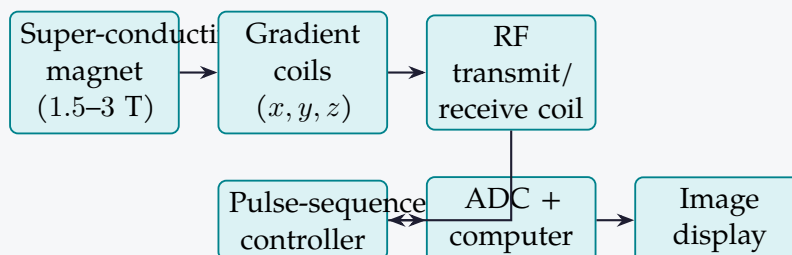
Spatial Encoding using Gradient Fields

A linear gradient $G_z = \partial B / \partial z$ makes the resonance frequency of a proton depend on its position z :

$$\nu(z) = \frac{\gamma}{2\pi} (B_0 + G_z z)$$

Combining x, y, z gradients with phase and frequency encoding produces a 2D or 3D image.

Major Components of an MRI Scanner



5.7 T_1 and T_2 Contrast

Image type	Contrast source	Tissue look
T_1 -weighted	Short TR, short TE; differences in spin-lattice relaxation	Fat is bright, water (CSF) is dark
T_2 -weighted	Long TR, long TE; differences in spin-spin relaxation	Water is bright, fat is intermediate
Proton density	Long TR, short TE	Reflects total ^1H count
fMRI (BOLD)	Magnetic susceptibility change of haemoglobin	Detects neural activity

5.8 Advantages and Limitations of MRI

Aspect	Advantages	Limitations
Radiation	Non-ionising, safe	—
Soft-tissue contrast	Excellent (best of all imaging modalities)	—
Cost	—	Very expensive scanner and cryogen
Acquisition time	—	Several minutes per sequence
Patient compatibility	—	Pacemakers, ferromagnetic implants, claustrophobia

5.9 Medical Applications of MRI

- Brain: tumour detection, multiple sclerosis, stroke (diffusion MRI).

- Spinal cord and joints: disc prolapse, ligament tears.
- Cardiac MRI: chamber size, ejection fraction, myocardial viability.
- Functional MRI (fMRI): mapping of brain activity using BOLD contrast.
- Magnetic resonance angiography (MRA) without iodinated contrast.
- Whole-body screening and oncology staging.

5.10 Origin of Chemical Shift: Shielding Mechanisms

A nucleus's resonance frequency is determined by the local field $B_{\text{eff}} = B_0(1 - \sigma)$. The shielding constant σ has three additive contributions in organic molecules.

Mechanism	Effect
Diamagnetic shielding (σ_d)	Local circulation of bonding electrons opposes B_0 . Increases with electron density on the nucleus. Electronegative neighbours <i>decrease</i> σ_d and shift signal downfield.
Paramagnetic shielding (σ_p)	Mixing of low-lying excited states with the ground state <i>adds</i> to B_{eff} and shifts signal downfield. Important in ^{13}C but small in ^1H .
Anisotropic / ring-current shielding	Aromatic π -electrons circulate in B_0 , generating a secondary field. Protons <i>above/below</i> the ring are shielded; protons <i>at the edge</i> are deshielded ($\delta \sim 7\text{--}8$ ppm).

Why TMS is the reference: Si is electropositive, the four equivalent CH_3 groups are heavily shielded (high electron density on H), and TMS has only one sharp singlet at very low δ , well outside the range of typical organic protons. It is also chemically inert, volatile, and miscible with most solvents.

5.11 Coupling Constant J and Integration

A ^1H NMR spectrum delivers four pieces of information; the multiplet pattern alone is not enough.

Information	Reads from
Chemical environment	Number of separate signals (peaks at different δ).
Chemical shift δ	Position in ppm; identifies the type of proton.
Coupling pattern	Number of neighbours through $n + 1$ rule.
Integration	Area under each peak; ratio gives <i>number of protons</i> in each environment.

Coupling Constant J

$$J \text{ (Hz)} = (\nu_2 - \nu_1)$$

J is measured directly from the line spacing within a multiplet and is *independent* of B_0 (chemical shift in Hz scales with B_0 , but J does not). This is the standard way to distinguish a closely-spaced multiplet from two coincidentally overlapping singlets.

Typical ^1H – ^1H coupling constants:

- Geminal (2J , on same C): -12 to $+15$ Hz.
- Vicinal (3J , on adjacent C, freely rotating): 6 – 8 Hz.
- Vicinal cis (alkene): 6 – 12 Hz.
- Vicinal trans (alkene): 12 – 18 Hz.
- Aromatic ortho: ~ 8 Hz, meta: ~ 2 Hz, para: < 1 Hz.

Solved Example: Distinguishing Cis and Trans Alkenes from J

Problem: A sample of *butenedioic acid* can be either maleic acid (cis) or fumaric acid (trans). The vinyl proton signals split into two doublets with $J = 11.6$ Hz.

Solution: J is in the cis range (6 – 12 Hz), well below the trans range (12 – 18 Hz). $\Rightarrow$ **maleic acid** (cis isomer).

5.12 MRI Contrast Agents

Some pathologies are hard to see in native T_1 or T_2 contrast. A paramagnetic contrast agent is then injected intravenously to enhance differences locally.

Class	Mechanism	Examples
Gadolinium chelates	Gd^{3+} has 7 unpaired electrons; shortens water T_1 , brightens regions where it accumulates	Gd-DTPA (Magnevist), Gd-DOTA (Dotarem)
SPIONs (super-paramagnetic iron oxide nanoparticles)	Strong local field inhomogeneities shorten T_2 , darken regions where they accumulate	Ferumoxytol, Resovist
Hyperpolarised gases / metabolites	External polarisation amplifies signal of ^{129}Xe , ^{13}C -pyruvate, etc.	Xe lung MRI, ^{13}C metabolic MRI

Safety note: Free Gd^{3+} is toxic; clinical agents use a strong chelator (DTPA, DOTA) to keep it bound until the agent is renally excreted. Patients with reduced kidney function ($\text{eGFR} < 30$) avoid Gd contrast due to nephrogenic systemic fibrosis risk.

Why MRI Has Better Soft-Tissue Contrast than X-ray CT

X-ray CT contrast comes only from electron density, so most soft tissues look similar. MRI exploits proton density *and* two relaxation times T_1, T_2 , giving multiple contrast knobs and clear discrimination of grey vs white matter, oedema vs tumour, etc.

Exam Points (Section 5)

- NMR-active nuclei need $I \neq 0$. List $^1\text{H}, ^{13}\text{C}, ^{19}\text{F}, ^{31}\text{P}$.
- Always quote Larmor equation $\nu = \gamma B_0 / 2\pi$.
- Use the $n + 1$ rule and Pascal's triangle to sketch multiplets.

Frequently Asked / Viva Questions (contd.)

- Define δ in ppm and reference TMS.
- For MRI, mention magnetic gradients for spatial encoding and T_1/T_2 contrast.
- Three shielding mechanisms: diamagnetic, paramagnetic, ring-current.
- Coupling constant J is in Hz and *independent* of B_0 (chemical shift in Hz is not).
- Distinguish cis/trans alkenes from J values: cis 6–12 Hz, trans 12–18 Hz.
- Gd-DTPA shortens water T_1 and brightens MRI images; SPIONs shorten T_2 and darken them.

6. Surface Characterisation Techniques

Surface characterisation techniques probe the outermost few atomic layers of a solid to give information about (i) topography (shape and roughness), (ii) elemental composition, (iii) chemical state of surface atoms, and (iv) electronic structure. They are essential for catalysts, semiconductors, thin films, biomaterials, and corrosion studies.

6.1 Why a Special Technique for Surfaces?

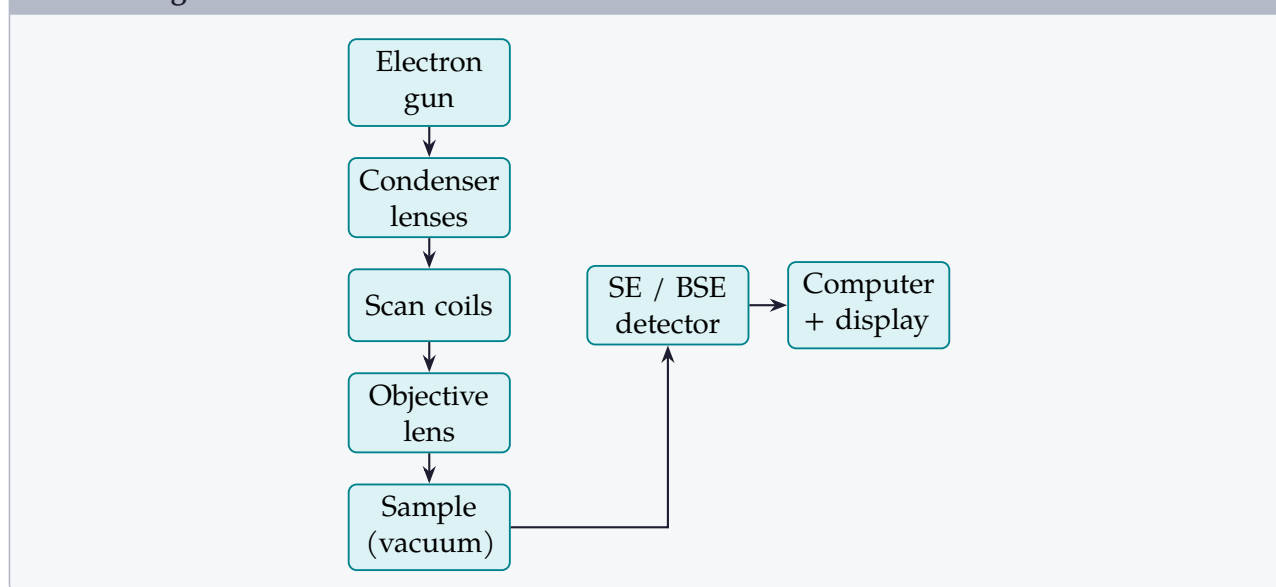
Bulk techniques (e.g. XRD) average over millions of atomic layers and miss surface effects such as adsorption, oxidation, segregation, and reconstruction. Surface techniques use probes (electrons, ions, photons, mechanical tip) that penetrate only a few nanometres.

Technique	Probe in	Signal out	Information
SEM	Electrons (5–30 kV)	Secondary / back-scattered electrons	Topography, morphology
TEM	Electrons (100–300 kV)	Transmitted electrons	Internal structure, lattice imaging
AFM	Sharp tip	Tip deflection / force	Atomic-scale topography
STM	Tunnelling current	Current vs z	Electronic + topographic image
XPS / ESCA	Soft X-rays	Photo-electrons (KE)	Surface composition, oxidation state
AES	Electron beam	Auger electrons	Elemental composition (light elements)
SIMS	Ion beam	Sputtered secondary ions	Trace elements, depth profile

6.2 Scanning Electron Microscopy (SEM)

A focused beam of electrons is rastered over the surface; secondary electrons (SE) reveal topography while back-scattered electrons (BSE) give compositional contrast.

Block Diagram of an SEM



Key features of SEM: resolution ~ 1 nm, depth of field very large (3D-looking images), needs conducting or coated sample, high vacuum.

6.3 X-ray Photoelectron Spectroscopy (XPS / ESCA)

A sample is irradiated with monochromatic soft X-rays ($\text{Al K}\alpha$, $h\nu = 1486.6$ eV). Core-level electrons are ejected; their kinetic energy is measured.

XPS Energy-Conservation Equation

$$E_B = h\nu - E_K - \phi$$

E_B = binding energy; E_K = kinetic energy of detected photo-electron; ϕ = spectrometer work function.

E_B is element-specific and shifts with chemical state, e.g. $\text{Fe}(0)$ vs Fe_2O_3 .

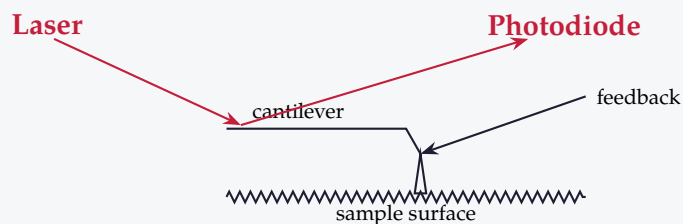
Information obtained from XPS:

- Elemental identification (all elements except H, He).
- Oxidation state and chemical environment (chemical shift).
- Quantitative composition of top 5–10 nm.
- Depth profiling when combined with Ar^+ sputtering.

6.4 Atomic Force Microscopy (AFM)

A nanometric tip on a flexible cantilever is scanned over the surface. A laser reflected off the cantilever measures sub-nanometre deflections that map height (topography) and tip–surface forces.

Working Principle of AFM



Modes of AFM:

- *Contact mode*: tip drags on surface; high resolution but can damage soft samples.
- *Tapping mode*: cantilever oscillates near resonance, tip touches lightly; safe for biology and polymers.
- *Non-contact mode*: tip senses long-range forces; lowest damage.

Advantages of AFM: works in air, liquid, and vacuum; no special coating; gives true 3D height; image biological molecules (DNA, proteins) directly.

6.5 Comparison of Major Surface Techniques

Feature	SEM	AFM	XPS
Probe	Electrons	Mechanical tip	Soft X-rays
Signal	SE / BSE	Force / deflection	Photo-electrons
Lateral resolution	~ 1 nm	~ 0.1 nm (atomic)	~ 10 μm (small spot) ~ 10 μm
Depth probed	Few nm (SE)	Top atomic layer	5–10 nm
Vacuum	High vacuum	Air / liquid OK	Ultra-high vacuum
Sample preparation	Coat if non-conductive	Almost none	Conductive desirable
Composition info	Possible with EDX	No (height only)	Yes, with chemical state

6.6 TEM vs SEM

The two electron-beam microscopies are complementary: SEM gives surface topography while TEM gives internal structure with atomic resolution.

Feature	SEM	TEM
Beam energy	5–30 kV	100–300 kV
What is detected	Secondary / back-scattered electrons <i>above</i> sample	Electrons <i>transmitted through</i> thin sample
Sample form	Bulk, polished, conductive coating if needed	Ultra-thin film (< 100 nm) on grid
Resolution	~ 1 nm	0.1–0.2 nm (atomic in HRTEM)
Image gives	Topography, depth of field, morphology	Internal structure, lattice fringes, defects
Diffraction option	Limited (EBSD)	Selected-area electron diffraction (SAED)

6.7 Scanning Tunnelling Microscopy (STM)

STM achieves true atomic resolution on conducting surfaces by exploiting the quantum tunnelling current that flows between a sharp metal tip and the surface when separated by ~ 1 nm.

Tunnelling Current in STM

$$I_T \propto V \exp(-2\kappa d), \quad \kappa = \sqrt{\frac{2m_e\phi}{\hbar^2}}$$

V = bias voltage; d = tip–surface gap; ϕ = work function. Because I_T falls off exponentially with d , a 0.1 nm change in d changes I_T by a factor of ~ 10 — that is the source of atomic-scale sensitivity.

STM operating modes:

- Constant-current mode: feedback adjusts z to keep I_T fixed; image is the $z(x, y)$ map.
- Constant-height mode: tip z is fixed; image is the $I_T(x, y)$ map; faster but only safe on flat samples.

Limitation: STM only works on conductors and semiconductors, while AFM works on insulators too. STM is the original “atomic resolution” technique that won Binnig and Rohrer the 1986 Nobel Prize.

6.8 BET Surface Area Analysis

The Brunauer–Emmett–Teller (BET) isotherm extends the Langmuir model to multilayer adsorption and lets you compute the specific surface area of a porous solid from N₂ adsorption at 77 K.

BET Equation

$$\frac{P/P_0}{V(1 - P/P_0)} = \frac{1}{V_m C} + \frac{C - 1}{V_m C} \left(\frac{P}{P_0} \right)$$

P/P_0 = relative pressure; V = adsorbed volume at P/P_0 ; V_m = monolayer volume; C = BET constant.

A plot of LHS vs P/P_0 is linear in the range 0.05–0.30. From the slope and intercept, V_m is obtained, and the specific surface area is

$$S_{\text{BET}} = \frac{V_m N_A a_m}{V_{\text{molar}}}$$

where $a_m \approx 0.162 \text{ nm}^2$ is the cross-sectional area of an N_2 molecule.

Why important: BET surface area is the standard metric for catalysts ($\text{Pt}/\gamma\text{-Al}_2\text{O}_3 \sim 200 \text{ m}^2/\text{g}$), zeolites ($\sim 600 \text{ m}^2/\text{g}$), activated carbon ($> 1000 \text{ m}^2/\text{g}$), and metal-organic frameworks (up to $7000 \text{ m}^2/\text{g}$).

6.9 Contact-Angle Measurement: Surface Wettability

A drop of water on a clean surface forms a contact angle θ with the surface. This single number is a powerful surface characterisation tool.

Young's Equation

$$\gamma_{SV} = \gamma_{SL} + \gamma_{LV} \cos \theta$$

$\gamma_{SV}, \gamma_{SL}, \gamma_{LV}$ = solid–vapour, solid–liquid, liquid–vapour interfacial tensions.

Hydrophilic surface: $\theta < 90^\circ$ (water spreads). **Hydrophobic** surface: $\theta > 90^\circ$ (water beads). **Superhydrophobic** (lotus leaf): $\theta > 150^\circ$.

Applications: surface treatment of biomaterials, anti-fouling coatings, microfluidic device design, paint-adhesion checks.

Exam Points (Section 6)

- Distinguish surface from bulk techniques and explain why probe penetration matters.
- For SEM: mention gun, lenses, scan coils, SE/BSE detectors, vacuum.
- Quote the XPS equation $E_B = h\nu - E_K - \phi$ and one application (oxidation state).
- Sketch AFM tip + cantilever + laser + photodiode.
- TEM gives internal structure (0.1 nm); SEM gives surface topography (1 nm).
- STM tunnelling current $I_T \propto \exp(-2\kappa d)$; works only on conductors.
- BET surface area uses N_2 adsorption at 77 K; activated carbon $> 1000 \text{ m}^2/\text{g}$.
- Contact angle: $< 90^\circ$ hydrophilic, $> 90^\circ$ hydrophobic, $> 150^\circ$ superhydrophobic.

7. Diffraction and Scattering

Diffraction is the constructive interference of waves scattered by a regular array of atoms or molecules. It produces sharp peaks and is the principal tool for determining crystal structure. **Scattering** more generally is the redirection of radiation by particles, with or without change of energy; it gives information about size, shape and electron distribution of the scatterers.

7.1 Bragg's Law of X-ray Diffraction

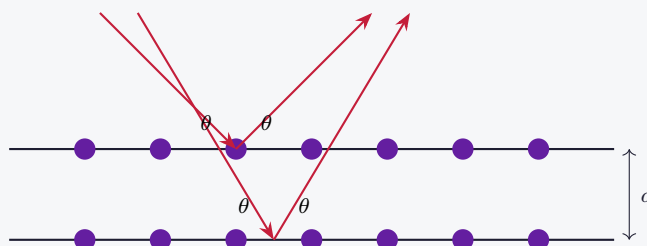
A crystal acts like a 3D grating because atoms lie on parallel planes separated by an interplanar spacing d . Constructive interference occurs at angles satisfying Bragg's law.

Bragg's Law

$$n\lambda = 2d \sin \theta$$

n = order (integer); λ = wavelength of X-ray (typically Cu $K\alpha$, 1.5418 Å); d = interplanar spacing; θ = Bragg angle (half of the diffraction angle 2θ).

Bragg Reflection from Two Lattice Planes



Path difference: The lower ray travels an extra $2d \sin \theta$. Constructive interference (bright spot) requires this to equal $n\lambda$.

Solved Example: Spacing of NaCl from Bragg Reflection

Problem: A first-order ($n = 1$) Bragg reflection from NaCl is observed at $\theta = 15.5^\circ$ with Cu $K\alpha$ X-rays ($\lambda = 0.154$ nm). Find the interplanar spacing.

Solution:

$$d = \frac{n\lambda}{2 \sin \theta} = \frac{1 \times 0.154}{2 \times \sin 15.5^\circ} = \frac{0.154}{2 \times 0.2672} = 0.288 \text{ nm}$$

This corresponds to the (200) planes of NaCl.

7.2 Powder X-ray Diffraction (PXRD)

A polycrystalline powder presents all crystallographic orientations to the beam. Each $\{hkl\}$ plane satisfying Bragg's law produces a cone of diffracted intensity, recorded as a series of peaks vs 2θ .

Information from a powder pattern:

- Identification of phases (matched against the ICDD database).

- Lattice parameters from peak positions.
- Crystallite size from peak broadening (Scherrer formula).
- Quantitative phase analysis (Rietveld refinement).

Scherrer Equation for Crystallite Size

$$D = \frac{K\lambda}{\beta \cos \theta}$$

D = average crystallite size; $K \approx 0.9$ (shape factor); β = full-width at half-maximum (FWHM) in radians; θ = Bragg angle.

7.3 Block Diagram of an X-ray Diffractometer

Block Diagram of a Powder Diffractometer



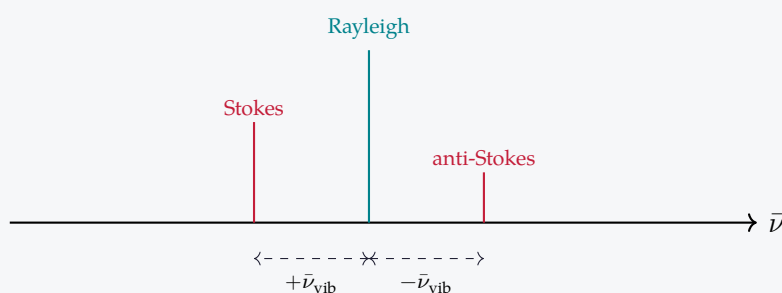
7.4 Electron and Neutron Diffraction

Probe	Wavelength	Speciality
X-ray	0.05–0.3 nm	Senses electron density; routine crystallography.
Electron	~ 0.005 nm at 100 kV	Strongly absorbed; ideal for thin films and TEM.
Neutron	0.1–0.3 nm	Senses nuclear positions, locates light atoms (H, Li); resolves magnetic structure.

7.5 Rayleigh and Raman Scattering

When light passes through a transparent medium, a small fraction is scattered. Most of the scattered photons retain the original frequency (*Rayleigh*, elastic). A tiny fraction exchanges energy with molecular vibrations (*Raman*, inelastic).

Rayleigh, Stokes and Anti-Stokes Lines in Raman



Raman selection rule: the vibration must change the molecular polarisability. Combined

with the IR rule (dipole change), Raman and IR are often *complementary*.

Rayleigh Intensity Dependence

$$I_{\text{scat}} \propto \frac{1}{\lambda^4}$$

This λ^{-4} dependence explains why the sky is blue (short wavelengths scatter more) and why sunsets are red.

7.6 Other Scattering Techniques

Technique	Information
Small-angle X-ray scattering (SAXS)	Particle size, shape, 1–100 nm aggregates, polymers.
Dynamic light scattering (DLS)	Hydrodynamic radius and diffusion of nanoparticles in solution.
Static light scattering (SLS)	Molar mass and radius of gyration of polymers (Zimm plot).
Surface-enhanced Raman (SERS)	Trace molecule detection on rough Ag/Au surfaces; single-molecule sensitivity.

7.7 Applications of Diffraction and Scattering

- Crystal-structure determination of metals, ceramics, drugs and proteins.
- Phase identification in alloys and minerals.
- Quality control of pharmaceuticals (polymorphs).
- Thin-film analysis (texture, residual stress) by glancing-incidence XRD.
- Particle-size and aggregation studies (DLS, SAXS).
- Vibrational fingerprinting using Raman; complements IR.

7.8 Three Classical X-ray Diffraction Methods

There are three historical setups, each suited to a different sample form.

Method	Sample	What is varied
Laue method	Single crystal, fixed orientation	Polychromatic X-rays (white beam); each plane satisfies Bragg at its own λ . Used to find crystal symmetry and orient single crystals.
Rotating-crystal method	Single crystal mounted on a goniometer	Crystal rotated; monochromatic λ . Records reflections from all planes during rotation.
Debye–Scherrer (powder) method	Polycrystalline powder	Random orientations present every plane to the beam; monochromatic λ . Produces concentric cones, recorded as a 2θ pattern. Most common in materials labs.

7.9 Raman Effect: Quantum Picture

Stokes and anti-Stokes lines arise because the molecule can either gain (anti-Stokes) or lose (Stokes) one vibrational quantum during the inelastic scattering event.

Stokes / Anti-Stokes Intensity Ratio

$$\frac{I_{\text{anti-Stokes}}}{I_{\text{Stokes}}} = \left(\frac{\bar{\nu}_0 + \bar{\nu}_v}{\bar{\nu}_0 - \bar{\nu}_v} \right)^4 \exp\left(-\frac{hc \bar{\nu}_v}{k_B T}\right)$$

At room temperature anti-Stokes lines are much weaker because few molecules occupy the $v = 1$ state. Heating the sample increases the anti-Stokes/Stokes ratio — an old way to measure local temperature.

7.10 Why the Sky is Blue: A Worked Rayleigh Calculation

Sunlight contains all visible wavelengths. Each colour scatters with intensity $\propto 1/\lambda^4$.

Solved Example: Blue / Red Scattering Ratio

Problem: Compare the Rayleigh-scattered intensity of blue ($\lambda = 470 \text{ nm}$) and red ($\lambda = 650 \text{ nm}$) light.

Solution:

$$\frac{I_{\text{blue}}}{I_{\text{red}}} = \left(\frac{650}{470} \right)^4 = (1.383)^4 = 3.66$$

Blue light is scattered $\sim 3.7\times$ more strongly than red. When you look up away from the sun, you see this scattered blue light; when you look directly at sunset/sunrise the direct beam has been depleted of blue and appears red.

7.11 Raman Spectroscopy: Practical Applications

Beyond the engineering chemistry textbook, Raman is now a workhorse technique in industry and forensics.

- Pharmaceutical quality control: through-bottle identification of API and excipients (TRS

technology).

- Polymorph screening in drug formulation.
- Forensic analysis of paints, fibres, narcotics — non-destructive, no sample prep.
- Identification of pigments in art and archaeology (no sampling of priceless objects).
- Carbon-material characterisation: D, G and 2D bands in graphene/CNT.
- Biomedical Raman: in-vivo cancer margin detection (handheld probes).
- SERS for trace explosives and pesticide residue detection.

IR vs Raman: Complementary Spectra of CO₂

CO₂ has four vibrational modes. The symmetric stretch is Raman-active (IR-inactive), while the asymmetric stretch and the two bends are IR-active (Raman-inactive). Together IR + Raman map all modes — a direct application of the mutual exclusion principle.

Exam Points (Section 7)

- Always state Bragg's law with all symbols defined.
- Mention the use of Cu K α ($\lambda = 1.54 \text{ \AA}$) and that we measure peak position 2θ .
- For nanoparticles, recall Scherrer's equation $D = K\lambda/(\beta \cos \theta)$.
- Distinguish Rayleigh (elastic) vs Raman (inelastic) and link to selection rules.
- Quote the $1/\lambda^4$ Rayleigh law and "why the sky is blue".
- Three classical XRD methods: Laue (single crystal, polychromatic), rotating-crystal, Debye-Scherrer (powder).
- Anti-Stokes/Stokes intensity ratio depends on temperature — thermometry application.
- SERS gives single-molecule sensitivity for forensic and trace detection.

Quick Revision — Module 2

Spectroscopic Techniques and Applications | BS-CH101

Key Formulas at a Glance

Formula / Rule	Meaning
$E = h\nu = \frac{hc}{\lambda} = hc\bar{\nu}$	Photon energy in terms of ν , λ , $\bar{\nu}$.
$\Delta E = h\nu$	Bohr resonance condition for absorption / emission.
$P_{1 \rightarrow 2} \propto \mu_{12} ^2$	Transition probability; selection rules emerge here.
$\frac{N_2}{N_1} = \frac{g_2}{g_1} e^{-\Delta E/k_B T}$	Boltzmann level population.
$A = \varepsilon cl$	Beer–Lambert law for UV-Vis quantitation.
$\Phi_F = \frac{k_F}{k_F + k_{NR} + k_{ISC}}$	Fluorescence quantum yield.
$\frac{F_0}{F} = 1 + K_{SV}[Q]$	Stern–Volmer quenching.
$\bar{B} = \frac{h}{8\pi^2 c I}, I = \mu r^2$	Rotational constant and moment of inertia.
$E_v = (v + \frac{1}{2})h\nu_0$	Harmonic-oscillator vibrational energy.
$\bar{\nu}_0 = \frac{1}{2\pi c} \sqrt{\frac{k}{\mu}}$	Vibrational wavenumber.
$\nu_L = \frac{\gamma B_0}{2\pi}$	NMR Larmor (resonance) frequency.
$\delta \text{ (ppm)} = \frac{\nu_{\text{sample}} - \nu_{\text{TMS}}}{\nu_{\text{spec}}} \times 10^6$	NMR chemical shift relative to TMS.

Key Formulas at a Glance (contd.)

$E_B = h\nu - E_K - \phi$	XPS binding-energy equation.
$n\lambda = 2d \sin \theta$	Bragg's law for X-ray diffraction.
$D = \frac{K\lambda}{\beta \cos \theta}$	Scherrer equation, crystallite size.
$I_{\text{scat}} \propto 1/\lambda^4$	Rayleigh scattering law.
$\lambda_{\text{max}} = \lambda_{\text{base}} + \sum \text{increments}$	Woodward–Fieser empirical prediction.
$E = \frac{R_0^6}{R_0^6 + r^6}$	FRET efficiency.
$\frac{\tilde{\nu}_D}{\tilde{\nu}_H} = \sqrt{\mu_H/\mu_D}$	Isotope shift ($\approx 1/\sqrt{2}$ for X–H/X–D).
$\Phi_F = k_F \tau_F$	Quantum-yield/lifetime relation.
$\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3}$	Einstein A/B coefficient ratio.
$I_T \propto \exp(-2\kappa d)$	STM tunnelling current (atomic resolution).
$\gamma_{SV} = \gamma_{SL} + \gamma_{LV} \cos \theta$	Young's equation for contact angle.
$S_{\text{BET}} = \frac{V_m N_A a_m}{V_{\text{molar}}}$	BET specific surface area.

One-Line Definitions

- **Spectroscopy:** Study of interaction of EM radiation with matter.
- **Selection rule:** Quantum-number condition that decides whether a transition is allowed.
- **Chromophore:** Functional group responsible for absorption of UV-Vis light.
- **Auxochrome:** Saturated group that shifts and intensifies an existing chromophore.
- **Bathochromic shift:** Movement of λ_{max} to longer wavelength (red shift).
- **Fluorescence:** Spin-allowed radiative emission from S_1 to S_0 .
- **Phosphorescence:** Radiative $T_1 \rightarrow S_0$ emission, long-lived afterglow.
- **Stokes shift:** Difference between absorption and emission maxima.
- **Quantum yield (Φ_F):** Fraction of absorbed photons re-emitted as fluorescence.
- **Rotational constant ($\bar{B}$):** Inverse moment of inertia in cm^{-1} .
- **Reduced mass (μ):** $m_1 m_2 / (m_1 + m_2)$ for a diatomic molecule.
- **Force constant (k):** Stiffness of a bond modelled as a spring.
- **Chemical shift (δ):** NMR resonance position relative to TMS, in ppm.
- **Larmor frequency:** Resonance ν of a nuclear spin in a field B_0 .
- **MRI gradient:** Linear field variation that encodes spatial position.

- **Bragg angle:** Half of the diffraction angle at which constructive interference occurs.
- **Crystallite size (Scherrer):** Average coherent domain size estimated from peak broadening.
- **Rayleigh scattering:** Elastic redirection of light by sub-wavelength particles.
- **Raman scattering:** Inelastic scattering with vibrational energy exchange.
- **Mutual exclusion principle:** For molecules with a centre of symmetry, IR-active modes are Raman-inactive and vice-versa.
- **Franck–Condon principle:** Electronic transitions are vertical because nuclei do not move on the timescale of electronic excitation.
- **Einstein coefficients:** Three rate constants A_{21}, B_{12}, B_{21} describing spontaneous emission, stimulated absorption and stimulated emission.
- **Kasha's rule:** Emission always originates from the lowest excited state of a given multiplicity.
- **FRET:** Distance-dependent non-radiative dipole–dipole energy transfer; $E \propto 1/r^6$.
- **Woodward–Fieser rules:** Empirical formula to predict $\lambda_{\max}$ of dienes and enones.
- **Isosbestic point:** Wavelength where two interconverting species have equal ε .
- **Coupling constant (J):** Spacing within a multiplet in Hz; independent of B_0 .
- **Integration:** Area under an NMR signal, proportional to the number of equivalent protons.
- **Ring-current effect:** Aromatic π -electron circulation deshields ring protons.
- **Gd-DTPA:** Gadolinium chelate used as T_1 -shortening MRI contrast agent.
- **BET surface area:** Specific surface area measured by N_2 adsorption at 77 K.
- **STM:** Atomic-resolution microscopy based on quantum tunnelling current between tip and conducting surface.
- **Contact angle:** Angle a liquid drop makes with a solid surface; quantifies wettability.
- **Laue method:** Single-crystal XRD with a polychromatic beam, used to determine crystal orientation.

Frequently Asked / Viva Questions

Q1. State the principle of spectroscopy.

A molecule absorbs a photon when its energy matches the gap between two allowed levels: $\Delta E = h\nu$.

Q2. What are gross and specific selection rules?

Gross rule sets a property the molecule must have (dipole, polarisability, spin), specific rule restricts the change in quantum numbers (e.g. $\Delta v = \pm 1$).

Frequently Asked / Viva Questions (contd.)

Q3. Why is N₂ IR-inactive but Raman-active?

The N₂ stretch does not change the dipole moment (zero) but does change polarisability.

Q4. State Beer–Lambert law and one limitation.

$A = \epsilon cl$. It fails at high concentration because intermolecular interactions alter ϵ .

Q5. Define chromophore and give two examples.

Unsaturated group responsible for UV-Vis absorption; e.g. C=O, C=C.

Q6. What is the difference between fluorescence and phosphorescence?

Fluorescence is spin-allowed $S_1 \rightarrow S_0$ (10^{-9} – 10^{-7} s); phosphorescence is spin-forbidden $T_1 \rightarrow S_0$ (10^{-3} – 10 s).

Q7. Name two medical uses of fluorescence.

Fluorescein retinal angiography and photodynamic therapy of cancer.

Q8. Why does HCl show a microwave spectrum but H₂ does not?

Only HCl has a permanent electric dipole; the rotational gross rule fails for H₂.

Q9. State the harmonic-oscillator selection rule for IR.

$\Delta v = \pm 1$, with the requirement that the vibration changes the dipole moment.

Q10. State the Larmor equation.

$\nu_L = \gamma B_0 / (2\pi)$; for ¹H at 11.74 T, $\nu_L = 500$ MHz.

Q11. What is chemical shift in NMR?

Position of a resonance in ppm relative to TMS: $\delta = (\nu_s - \nu_{\text{TMS}}) / \nu_{\text{spec}} \times 10^6$.

Q12. How does MRI distinguish tissues?

Through differences in proton density, T_1 and T_2 relaxation times of water in each tissue.

Q13. Give two surface techniques and one piece of information each.

SEM (topography), XPS (oxidation state of surface atoms).

Q14. State Bragg's law.

$n\lambda = 2d \sin \theta$; gives d -spacings of crystal planes from peak positions.

Q15. Differentiate Rayleigh and Raman scattering.

Rayleigh is elastic, same ν . Raman is inelastic, $\pm \bar{\nu}_{\text{vib}}$ shifted lines.

Q16. State Franck–Condon principle.

Electronic transitions are vertical on a potential-energy diagram because nuclei do not move during the 10^{-15} s duration of electronic absorption.

Frequently Asked / Viva Questions (contd.)

Q17. What does Kasha's rule state?

Emission almost always originates from the lowest vibrational level of the lowest excited state of a given spin multiplicity, regardless of the excitation wavelength.

Q18. Define FRET and give one application.

Non-radiative resonance energy transfer from an excited donor to a nearby acceptor; $E \propto R_0^6/(R_0^6 + r^6)$. Used as a molecular ruler in single-molecule biophysics and TaqMan PCR.

Q19. State Woodward–Fieser rules.

Empirical addition of base value plus increments for substituents predicts $\lambda_{\max}$ of dienes (215 nm acyclic) and α, β -unsaturated ketones (215 nm acyclic) within ± 5 nm.

Q20. What is an isosbestic point?

Wavelength where two interconverting species have equal ϵ ; presence of a sharp isosbestic point confirms a clean two-component equilibrium.

Q21. Why does D–Cl absorb at lower wavenumber than H–Cl?

Same force constant but doubled reduced mass; $\bar{\nu} \propto 1/\sqrt{\mu}$ gives a factor of ~ 0.71 . So DCl absorbs at $\sim 2137 \text{ cm}^{-1}$ vs HCl at 2989 cm^{-1} .

Q22. What information does NMR integration give?

Area under a peak is proportional to the number of equivalent protons in that environment, providing the H-count of each chemical type.

Q23. What is the coupling constant J and how do you measure it?

Spacing within a multiplet, expressed in Hz, independent of B_0 . It is read directly from line spacing in the spectrum and is diagnostic of cis/trans, geminal/vicinal coupling.

Q24. Name two MRI contrast agents.

Gd-DTPA (Magnevist) shortens water T_1 to brighten tissues; SPIONs shorten T_2 to darken regions where they accumulate.

Q25. How does STM achieve atomic resolution?

Tunnelling current $I_T \propto \exp(-2\kappa d)$ falls off exponentially with tip–surface gap, so a 0.1 nm change alters the current by $\sim 10\times$.

Q26. What is BET surface area?

Specific surface area of porous solids measured from N_2 adsorption at 77 K using the BET multilayer-isotherm equation. Activated carbon $> 1000 \text{ m}^2/\text{g}$.

Q27. What is the contact angle and what does it tell you?

Angle a sessile drop makes with a surface; $< 90^\circ$ hydrophilic, $> 90^\circ$ hydrophobic, $> 150^\circ$ superhydrophobic. Quantifies surface wettability via Young's equation.

Frequently Asked / Viva Questions (contd.)

Q28. Why is the sky blue and the sunset red?

Rayleigh-scattered intensity $\propto 1/\lambda^4$, so blue scatters $\sim 3.7\times$ more than red. Looking up reveals scattered blue; the direct sunset beam has been depleted of blue and looks red.

Summary Reference Table of Spectroscopic Techniques

Technique	Region / probe	Information	Key formula / rule
UV-Vis	200–800 nm photons	Electronic transitions, concentration	$A = \varepsilon cl$
Fluorescence	UV-Vis emission	Excited state, medical imaging	Φ_F , Stern–Volmer
IR / FT-IR	400–4000 cm^{-1}	Functional groups	$\bar{\nu}_0 = \frac{1}{2\pi c} \sqrt{k/\mu}$
Microwave	1–300 GHz	Bond length, I , $\bar{B}$	$\bar{B} = h/(8\pi^2 cI)$
NMR	RF in strong B_0	H/C environments, structure	$\nu_L = \gamma B_0/(2\pi)$
MRI	RF + magnetic gradients	Soft-tissue images	T_1, T_2 contrast
SEM / AFM / XPS	Electrons / tip / X-rays	Surface morphology + chemistry	$E_B = h\nu - E_K - \phi$ (XPS)
XRD	X-rays on crystal	Crystal structure, d -spacings	$n\lambda = 2d \sin \theta$
Raman	Vis/NIR scattering	Vibrational + symmetry info	Polarisability change